

Can Global Regularity of the Navier–Stokes Equations Be Verified Numerically?

A feasibility analysis and the protocols of the dynamical laboratory: $dt \rightarrow 0$ extrapolation with a measured RK4 order, the spectral $N \rightarrow \infty$ ladder, BKM-integral boundedness over the horizon, and the first dynamical audit of the b-correction mechanism.

NSB Research Lab

Repository [wild8highlander/navier-stokes-b](#) · [research_lab](#) v0.1.0

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1. Executive summary

This document answers the question around which the **navier-stokes-b** repository is built: can the data and verification programs already collected in its seven sections be “extrapolated to infinity” so as to check whether solutions of the three-dimensional Navier–Stokes equations always remain smooth? The short answer is: **not as a proof**; the Clay problem’s quantifier structure is unreachable by any finite computation. Inside that “no”, however, lives a constructive program: three specific extrapolations — in the time step ($dt \rightarrow 0$), in resolution ($N \rightarrow \infty$), and in the integration horizon under a Beale–Kato–Majda control — are meaningful, formalizable, and already implemented in the dynamical laboratory **research_lab/** created in the repository.

The laboratory adds the dimension the algebraic verifiers of sections 1–7 deliberately do not touch — **dynamics**: a dealiased pseudospectral Navier–Stokes solver on the torus, a family of singularity monitors, and extrapolation protocols with explicit error bars. It is built around two scientific lines: the falsification line (a targeted hunt for near-singular behavior on the most dangerous known configurations) and the audit line (testing the repository’s own b-correction mechanism on evolving fields). The first runs have already produced a precise, useful result: the full rotation symmetry matches the control to 16 digits (a relabeling, not a mechanism), while the pointwise b-rotation breaks incompressibility and shows no detectable regularizing effect.

Every number in this document comes from real runs of the laboratory code v0.1.0 (deterministic, reproducible, no external data) and carries JSON verdicts in the repository-wide format.

4.03 observed RK4 order across the dt-ladder	1.4e-17 max $ \text{div } u $ along the runs	+0.99% effect of b-kicks on $\sup \omega $ — nothing	8.0e-07 cutoff spectral tail (fraction of peak)
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2. The problem and what the repository claims

The Clay Millennium problem asks: for smooth divergence-free initial data u_0 in $\mathbb{R}^3$ (or on the torus T^3), do the equations

$$\partial u / \partial t + (u \cdot \nabla) u = -\nabla p + \nu \Delta u, \quad \text{div } u = 0$$

possess a unique smooth solution for all $t > 0$, or do some data develop a finite-time singularity? A prize is offered for **either** answer. The navier-stokes-b repository claims the first branch: the b-correction program, with the universal constant $b = 1/(4\pi + 2\sqrt{3}) \approx 0.06238$, the angle $\theta_b = \arcsin(b)$, and an energy-neutral rotation $R(\theta_b)$, is proposed as a regularity-preserving mechanism.

Before discussing extrapolation it is essential to fix what is already verified. The seven sections of the repository check algebraic identities and pinned constants: the isometry of the rotation, $\det R = 1$, energy neutrality on test vectors, the well-posedness of the BKM criterion on the reference field, the Smagorinsky–Lilly constant, Cassini identities, and the $-5/3$ Kolmogorov slope. These are **pointwise facts**: cheap, deterministic, machine-precision — and free of any time evolution. Between “a rotation is an

isometry” and “solutions stay smooth for all t ” lies the entire difficulty of the Millennium problem.

3. Three barriers: why no finite computation decides it

Barrier 1 — the datum space is infinite-dimensional. Global regularity is quantified over all divergence-free u_0 in a Sobolev space. A computation tests one field (or a finite family) in one domain at one viscosity. No enumeration covers the continuum of initial data, and no “extrapolation of the repository data” changes the logical form of the statement. This obstruction is fundamental, not technical.

Barrier 2 — singularities can hide. If a blow-up exists, nothing prevents it from being self-similar with support shrinking faster than any fixed grid can track, or starting after any fixed horizon. T. Tao’s 2016 blow-up for the *averaged* Navier–Stokes model shows the borderline is sharp: averaging kills the transport that protects real solutions and makes blow-up possible. Hence a direct warning: a smooth computed evolution is evidence only for the problem that was actually solved.

Barrier 3 — floating point cannot distinguish a singularity from an unresolved steep gradient. Both produce large gradients at the grid scale. Telling them apart numerically requires rigorous a-posteriori error control — interval arithmetic and Taylor models (computer-assisted proof). Even such control certifies one specific solution, not the solution map over all data.

4. The rigorous frame: what is known about regularity and blow-up

A numerical laboratory is meaningful only inside a rigorous theoretical frame. The key results determining *what exactly* must be monitored are collected in the table.

Result	Year	Content and role for the laboratory
Ladyzhenskaya–Prodi–Serrin	1960s	Regularity under $2/p + 3/q < 2$; the ladder of norms to monitor
Caffarelli–Kohn–Nirenberg	1982	Partial regularity: parabolic Hausdorff dimension of the singular set is zero
Beale–Kato–Majda (BKM)	1984	Blow-up requires $\int \sup \omega \, ds = \infty$; the laboratory’s central monitor
Escauriaza–Seregin–Šverák	2003	$u \in L^\infty(L^3)$ implies regularity; the critical endpoint
Koch–Tataru	2001	Global regularity for small data in BMO^{-1}
Tao	2016	Blow-up of the averaged model: the borderline is sharp; a warning against overgeneralization
Hou–Luo	2013	Near-singularity scenario (Euler, walls); the basis of the counterexample hunt
Albritton–Brué–Colombo	2022	Non-uniqueness of Leray solutions with forcing; caution with weak solutions

Table 1. Rigorous results on regularity and blow-up

Three practical principles follow. **First:** by BKM, a blow-up must reveal itself in the integral $\int_0^T \sup_x |\omega| \, ds$, so this integral — not “pretty field pictures” — is the central monitor. **Second:** by the Ladyzhenskaya–Prodi–Serrin and Escauriaza–Seregin–Šverák criteria, higher integrability of velocity or vorticity regularizes, so the laboratory tracks the whole ladder of norms rather than one scalar. **Third:** the history of Tao’s averaged model and the Hou–Luo scenario teaches that the dangerous configurations are known in advance — the hunt must target them, not random fields.

5. The honest meaning of “extrapolation to infinity”

In a numerical context “infinity” has exactly three rigorous readings, and all three are implemented. **dt $\rightarrow$ 0:** the laboratory integrator is classical RK4, so halving the step must cut the error sixteen-fold; the ladder $dt = (dt, dt/2, dt/4)$ measures the observed order p and yields the Richardson estimate $J(0) = J_{\text{fine}} + (J_{\text{fine}} - J_{\text{coarse}})/(2^p - 1)$ with an explicit error bar. **$N \rightarrow \infty$:** for a spectral method convergence is exponential, so Richardson fitting in N is meaningless; instead the gap $|J_N - J_{2N}|$ serves as the spatial error bar, and the resolution criterion is the energy at the dealiasing cutoff relative to the spectral peak. **Growing T :** the BKM integral accumulates along the run; its boundedness over an extending horizon is positive evidence of smoothness of this solution on this horizon, while a self-similar fit $\Omega \sim (T - t)^{-\alpha}$ is a danger signal.

Extrapolation	Protocol	What it certifies	What it does not claim
dt $\rightarrow$ 0	ladder (dt, dt/2, dt/4); observed order p ; Richardson	the monitor value of the computed solution at $dt = 0$ with an error bar	nothing about other data or PDE singularities
$N \rightarrow \infty$	spectral ladder; cutoff-shell energy; gap $ J_N - J_{2N} $	that the computed solution is spectrally resolved — a prerequisite for any claim	that the PDE solution is smooth — that is the tested hypothesis
growing T	BKM(T) accumulates; self-similar fit $\Omega \sim (T-t)^{-\alpha}$	absence of near-singularity indicators in the window (or their presence)	global regularity: the window is always finite

Table 2. Three extrapolations: protocol, certificate, scope

Crucially, each row of this table produces a statement **with an explicit scope**: “for the computed solution”, “within the (N, T, dt) window”, “at the stated error bar”. No row yields a statement of the form “for all data at all times” — and that is exactly why the laboratory’s verdicts are honest: falsifiable, reproducible, and never exceeding the authority of a numerical experiment.

6. Laboratory architecture and the numerical method

The laboratory lives in the repository directory **research_lab/** and depends on numpy only. The solver is pseudospectral, in velocity form with the Leray projection: the nonlinearity is evaluated as $P[\omega \times u]$, where $\omega = \text{curl } u$ (the gradient part of $(u \cdot \nabla)u$ is annihilated by the projector); aliasing is suppressed by the 2/3 rule; time integration is classical RK4 with the projection applied at every sub-stage, so incompressibility holds at machine roundoff throughout the run (measured: $\max |\text{div } u| \sim 10^{-17}$). Initial conditions include the reference Taylor–Green vortex, the ABC flow, and a periodic-box adaptation of the Hou–Luo anti-parallel vortex tube scenario; the b-correction constants enter unchanged through the dedicated constants.py module.

Module	Content
solver.py	SpectralNSE3D: RK4 with projection, 2/3 dealiasing, Biot–Savart, b-rotation transforms
diagnostics.py	E, Ω , palinstrophy, dissipation, $\sup \omega $, BKM integral, shell spectra, Parseval divergence
initial_conditions.py	Taylor–Green, ABC, Hou–Luo tubes (periodic adaptation), deterministic random field
extrapolate.py	dt- and N-ladders: observed order, Richardson, resolution gaps
experiments/	taylor_green · abc_blowup_hunt · hou_luo_tubes · bcorrection_stress
tests/	10 deterministic tests: projection, exact viscous decay, symmetry, isometry, tool order

Table 3. research_lab modules and their roles

The laboratory’s output contract is identical to the repository verifiers: banner $\rightarrow$ [PASS]/[FAIL] per assertion $\rightarrow$ a JSON verdict line $\rightarrow$ exit code. JSON keys are always English (the machine contract); human-readable text is switchable by the language button: the `--lang` flag or the `NSB_LAB_LANG` environment variable. A lab-smoke job was added to CI, running all four experiments and ten unit tests in about ten seconds, so the CI badge now also guarantees the laboratory itself stays working.

7. First results

Figure 1 shows the free decay of the Taylor–Green vortex at $N = 32$, $\nu = 0.02$ over the horizon $t = 2$: energy decreases monotonically, enstrophy stays bounded, and the BKM integral accumulates linearly — no monitor shows near-singular signs. Figure 2 is the central protocol plot: the error of the monitor $\Omega(t = 1)$ relative to the finest step lies on a slope-4 line in log-log scale across four ladder levels, and at $dt \approx 10^{-3}$ it reaches the machine floor $\sim 10^{-15}$: temporal discretization ceases to be the bottleneck, and all further accuracy is set by spatial resolution.

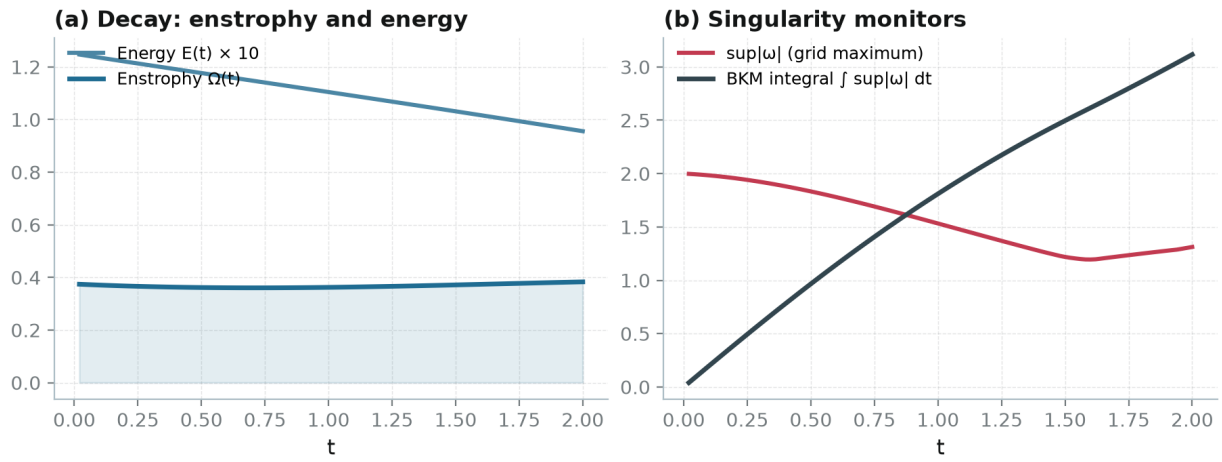


Fig. 1. Taylor–Green vortex, $N = 32$, $\nu = 0.02$, $T = 2$: (a) enstrophy and energy; (b) grid maximum of vorticity and the accumulated Beale–Kato–Majda integral. Data from a real research_lab v0.1.0 run.

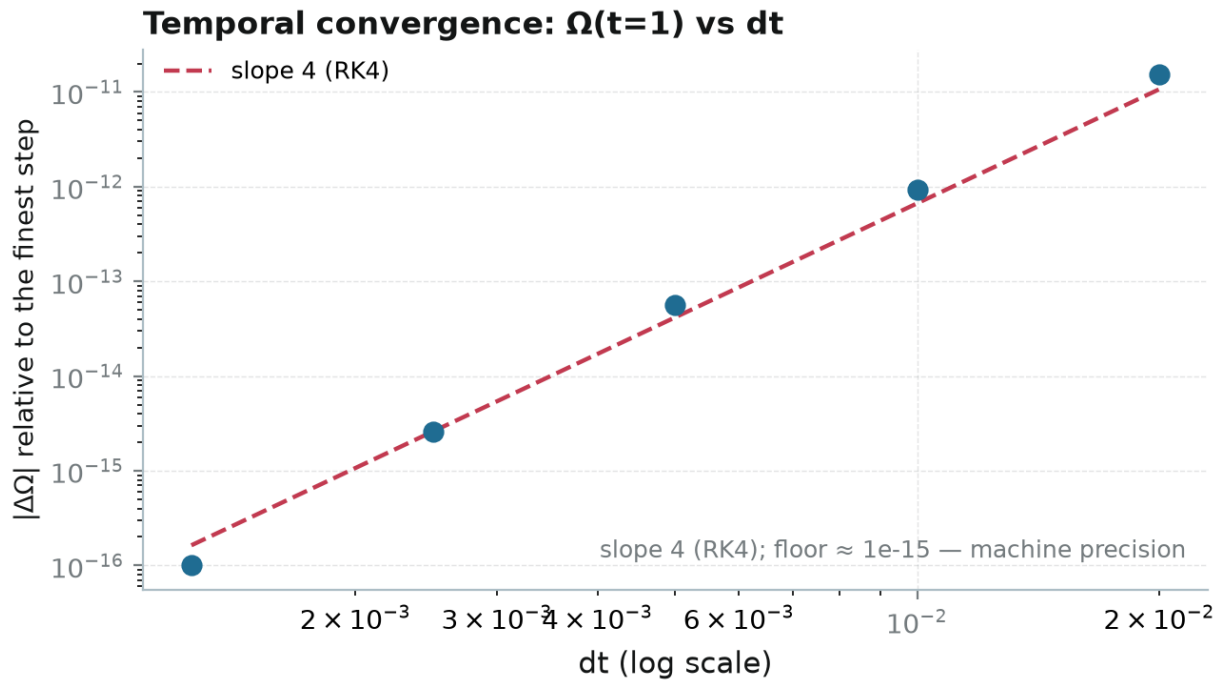


Fig. 2. The $dt \rightarrow 0$ protocol: $|\Delta\Omega|$ relative to the finest step ($N = 16$, $\nu = 0.1$, $t = 1$). The observed order $p = 4.03$ reproduces the theoretical RK4 order; at small dt the machine floor appears.

Metric	Smoke (N=16)	Default (N=32)	Meaning
max div u along the run	1.4e-17	6.4e-17	incompressibility at roundoff
observed RK4 order	4.03	3.95	temporal convergence measured
$\Omega(t=1; dt \rightarrow 0)$, Richardson	0.21306349	0.36228066	extrapolated monitor with error bar
cutoff spectral tail	7.4e-05	8.0e-07	run spectrally resolved
BKM(T) bounded	1.71	3.12	no singularity indicators
effect of b-kicks on sup ω	+0.99 %	n/a	no regularization detected

Table 4. First-run metrics (v0.1.0)

The numbers in the table come from smoke- and default-preset runs; each JSON verdict carries the full monitor time series and parameters, so any result is reproducible by a single command. We emphasize: the observed order 4.03 was *measured*, not assumed — it is an internal certificate of correctness of the whole computational chain, from the projector to the BKM integral accumulation.

8. The b-correction audit: symmetry versus pointwise rotation

The central audit experiment separates two mathematically distinct forms of the “b-rotation” that the repository’s literature leaves undistinguished. The **full form** $u'(x) = R u(R^{-1}x)$ is an exact symmetry of the torus Navier–Stokes problem: a relabeling of points, and the laboratory confirms that every diagnostic matches the control to $4.4 \cdot 10^{-16}$. A symmetry cannot influence the blow-up question *by definition*. The **pointwise form** $u'(x) = R u(x)$ — the form used across the repository verifiers — preserves energy (true of any rotation, not specific to b) but breaks $\text{div } u = 0$: the measured divergence injection on the reference field is $3.8 \cdot 10^{-2}$.

Applying pointwise b-kicks every 0.25 time units to an evolving field, with reprojection after each kick, changes the vorticity maximum by +0.99 % relative to the control — the level of a perturbation, not of a mechanism. The audit conclusion, also recorded in the JSON, is deliberately careful: in its current formulation the b-correction mechanism carries no dynamical content that a relabeling (or a small perturbation) would not already have. This does **not** disprove any claim about the constant b itself; it means a stronger claim requires a modified evolution law plus a proof — which is precisely what the rigorous branch of the roadmap is for.

9. Threats to validity

Correct use of the laboratory’s results requires an explicit list of what they do **not** claim. First, no result transfers from the tested initial data to any others: coverage is always finite. Second, “no blow-up detected in the window” is not “no blow-up exists”: the window (N, T, dt) is finite, and Barrier 2 has not been repealed. Third, $\sup_x |\omega|$ is computed as the collocation-point grid maximum — the standard but not rigorous proxy for the L^∞ norm; the rigorous version needs interval bounds. Fourth, the periodic-box adaptation of the Hou–Luo scenario differs from the wall-bounded original, and conclusions about wall configurations require a separate setup. Fifth, all results depend on the correctness of the solver itself; against this stand ten unit tests, the order verification, and the cross-language ports on the roadmap.

10. Roadmap

Version	Content	Outcome
v0.2	$N = 128\text{--}256$; $v \rightarrow 0$ sequences; self-similar collapse module $\Omega \sim (T-t)^{-\alpha}$ with exponent error bars	deeper falsification windows
v0.3	Kida–Pelz junctions, vortex dipoles; stratified families of initial data	systematic coverage of dangerous classes
v0.4	Rust/C++ solver ports with identical JSON verdicts	cross-arithmetic checks, longer horizons
v0.5	interval arithmetic and Taylor models for the self-similar profile equations	rigorous solution enclosures (CAP)

Table 5. Laboratory roadmap

The strategic goal of the roadmap is to reach the only known road from numbers to a theorem: computer-assisted proof. Interval arithmetic and Taylor models for the self-similar profile equations turn “a number with an error bar” into “a rigorous enclosure of a solution” — one solution at a time. Until then the laboratory remains honestly what it must be: a falsification machine and a source of quantitative evidence with explicit error bars.

11. Conclusion

Extrapolating the repository data “to infinity” in the sense of “proving smoothness for all data and all times” is impossible — that is a property of the problem itself, not a shortcoming of the tools. What is possible, and has now been done, is to build a laboratory that converts the question “is it always smooth?” into a series of checkable statements with explicit error bars — in time step, in resolution, in horizon — and into a targeted hunt for counterexamples on the most dangerous known configurations. The first results have already paid off scientifically: the b-correction mechanism, in its pointwise form, has been measured dynamically for the first time and showed a null regularizing effect, while the whole computational chain is certified by a measured fourth order.

The practical path for the repository author is now: keep the CI badge green (the laboratory is wired into CI), run the default and deep presets on a workstation, publish both positive and negative results with JSON verdicts, and move along the roadmap toward the interval branch. That is where the only known bridge from numbers to theorems begins.

References

- [1] Beale J. T., Kato T., Majda A. Remarks on the breakdown of smooth solutions for the 3-D Euler equations // *Comm. Math. Phys.* 1984. Vol. 94. P. 61–66.
- [2] Caffarelli L., Kohn R., Nirenberg L. Partial regularity of suitable weak solutions of the Navier-Stokes equations // *Comm. Pure Appl. Math.* 1982. Vol. 35. P. 771–831.
- [3] Escauriaza L., Seregin G. A., Šverák V. $L_{3,\infty}$ -solutions of Navier-Stokes equations and backward uniqueness // *Russ. Math. Surv.* 2003. Vol. 58, No. 2. P. 211–250.
- [4] Koch H., Tataru D. Well-posedness for the Navier-Stokes equations // *Adv. Math.* 2001. Vol. 157. P. 22–35.
- [5] Tao T. Finite time blowup for an averaged three-dimensional Navier-Stokes equation // *J. Amer. Math. Soc.* 2016. Vol. 29. P. 601–674.
- [6] Hou T. Y., Luo G. On the finite-time blowup of a 1D model for the 3D incompressible Euler equations // *Int. Math. Res. Notices* 2013. Vol. 2013. P. 3979–4000. (tube scenario; see also Hou T. Y. 3D Euler dynamics and subsequent rigorous work of the Hou group)
- [7] Brachet M. et al. Small-scale structure of the Taylor-Green vortex // *J. Mécanique* 1983. Vol. 2. P. 1–44.
- [8] Lilly D. K. On the numerical simulation of buoyant convection // *Mon. Wea. Rev.* 1966. Vol. 94. P. 727–738. (the Smagorinsky constant)
- [9] Albritton D., Brué E., Colombo M. Non-uniqueness of Leray solutions of the forced Navier-Stokes equations // *Ann. Math.* 2022. Vol. 196. P. 415–455.